1. Basic OOP – Class and Object

```
class Student:
    def display(self):
        print("Welcome to Python OOP")


s1 = Student()
s1.display()
```

Output

Welcome to Python OOP

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2. Basic OOP – Student Details

```
class Student:
    def get_details(self, name, age):
        print("Name:", name)
        print("Age:", age)


s1 = Student()
s1.get_details("Athar", 30)
```

Output

Name: Athar
Age: 30

---

3. Constructor Example

```
class Student:
    def __init__(self):
        print("Constructor is called")


s1 = Student()
```

Output

Constructor is called

---

4. Constructor with Parameters

```
class Student:
    def __init__(self, name, age):
        self.name = name
        self.age = age

    def display(self):
        print("Name:", self.name)
        print("Age:", self.age)
```

```
s1 = Student("Athar", 30)
s1.display()
```

Output

```
Name: Athar
Age: 30
```

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5. Inheritance Example

```
class Animal:
    def eat(self):
        print("Animal is eating")


class Dog(Animal):
    def bark(self):
        print("Dog is barking")


d = Dog()

d.eat()
d.bark()
```

Output

```
Animal is eating
Dog is barking
```

---

6. Constructor with Inheritance

```
class Person:
    def __init__(self, name):
        self.name = name


class Employee(Person):
    def display(self):
        print("Employee Name:", self.name)


e = Employee("Athar")
e.display()
```

Output

```
Employee Name: Athar
```